

Weakly Supervised Localization of Otolith Growth Increments

A Technical Report on Density Head Training with DINO and CORAL

Written in ASD-STE100 style

1. Purpose

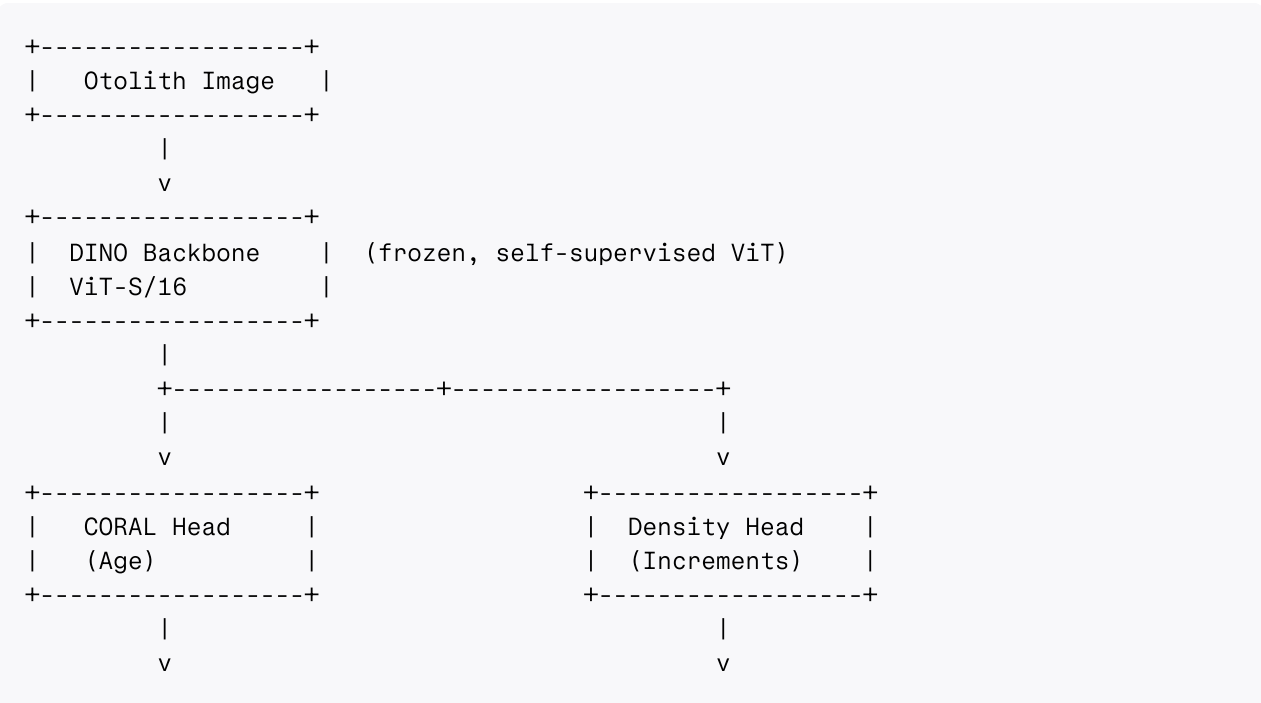
This report explains how to find growth increments on herring otolith images. The system uses a DINO Vision Transformer backbone. The system has two heads. The first head predicts the age of the otolith. The second head makes a density map. The density map shows where the growth increments are.

The report reviews the literature that supports this approach. The report describes eight method variants. The report also explains how to test the density head. A small set of expert-annotated images makes this test possible.

2. System Concept

2.1 Architecture Overview

The system has three main parts. The parts are the backbone, the age head, and the density head.



```

+-----+
| Age Prediction |
| (ordinal)      |
+-----+

```

```

+-----+
| Density Map    |
| (heat map)     |
+-----+

```

2.2 Term Definitions

Term	Definition
DINO backbone	A Vision Transformer trained without labels. It makes features that show object parts.
CORAL head	A head for ordinal regression. It predicts the age as a rank-consistent value.
Density head	A head that makes a spatial map. The map shows where growth increments are.
Expert points	Points marked by a human expert. Each point shows one growth increment.
Reference density map	A map made from expert points. It is the target for the density head.
Weak supervision	Training with labels at the image level only. No pixel-level labels are used.
Count-level supervision	Training with the total count only. The count is the age of the otolith.

2.3 Data Flow

```

Expert points (x, y)      Age label (integer)
      |                      |
      v                      |
+-----+                  |
| Make Gaussian            |
| density map              |
+-----+                  |
      |                      |
      v                      v
+-----+-----+
| Training loop            |
|                          |
| Loss = CORAL_loss + Density_loss |
+-----+-----+

```

3. Why the Small Annotated Set Matters

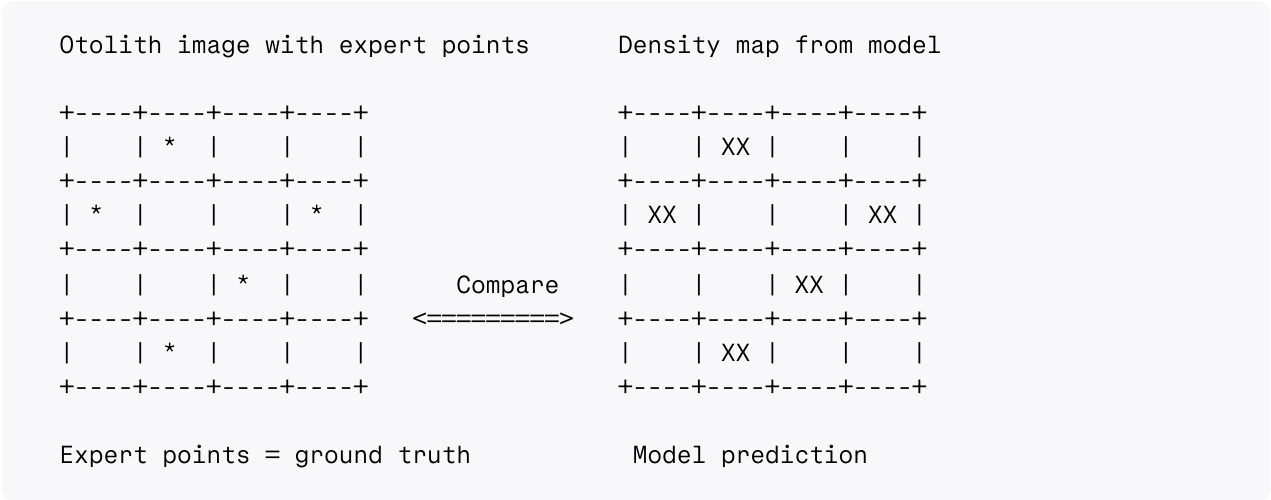
3.1 The Problem with Weak Supervision

Weak supervision trains a model with image-level labels only. The model learns to predict the age. But weak supervision does not tell the model where the increments are. The model must learn the locations by itself.

This creates a problem. You cannot know if the density map is correct. You can check the total count. You can check the age prediction. But you cannot check the spatial accuracy.

3.2 The Solution: Expert Annotations

A small set of expert-annotated images solves this problem. An expert marks the growth increments on each image. These marks are the ground truth. You can compare the density map to the expert marks.



3.3 What the Expert Set Lets You Measure

The expert-annotated set lets you compute these metrics:

Metric	What it measures	How to compute it
Count error	Does the density map sum to the correct age?	<code>abs(sum(density_map) - true_age)</code>
Point distance	How far are the predicted peaks from the expert points?	Mean Euclidean distance to nearest expert point
IoU	Do the predicted regions match the expert regions?	Intersection over union after thresholding
Precision	How many predicted increments are correct?	True positives / all predictions
Recall	How many expert increments did the model find?	True positives / all expert points
F1 score	Balance between precision and recall	$2 * P * R / (P + R)$
AUPRC	Area under the precision-recall curve	Integral of the PR curve

3.4 Two Data Sets

+-----+-----+-----+-----+ Set A: Annotated (small) - 20 to 50 images - Expert points on each image - Age label - Used for: * Direct density supervision * Localization evaluation	+-----+-----+-----+-----+ Set B: Age-only (large) - 1000+ images - Age label only - No increment locations - Used for: * Count-level supervision * CORAL training
---	---

	* Ablation study			* Prevent overfitting	
+	-----	+	+	-----	+

3.5 Ablation Study: How Small Can Set A Be?

The ablation study tests different sizes of Set A. It answers a key question. How many annotated images does the model need?

Number of annotated images	Expected result
-----	-----
5 images	May fail to generalize
10 images	Marginal localization
20 images	Acceptable localization
30 images	Good localization
50 images	Strong localization
100 images	Reference (upper bound)

4. Literature Evidence

This section reviews the literature. Each subsection covers one method family.

4.1 DINO: Self-Supervised Vision Transformers

Paper: Caron et al. (2021). "Emerging Properties in Self-Supervised Vision Transformers."

Link: <https://arxiv.org/abs/2104.14294>

DINO trains a Vision Transformer without labels. The method uses self-distillation. The model matches its own outputs across different views of the same image.

The key finding is this. The attention maps from DINO show object-like regions. This happens without any labels. The features also support good classification with a simple nearest-neighbor method.

Relevance: DINO attention maps may show growth increment regions on otolith images. The backbone does not need otolith-specific pretraining. The features are general and structural.

4.2 Deep ViT Features as Dense Visual Descriptors

Paper: Amir et al. "Deep ViT Features as Dense Visual Descriptors."

Link: <https://dino-vit-features.github.io/paper.pdf>

This work shows that DINO ViT features work as dense descriptors. The features support zero-shot part co-segmentation. They also support semantic correspondence finding.

Relevance: The patch-level features from DINO contain spatial information. This information can locate structures on the otolith surface.

4.3 TokenCut: Unsupervised Object Discovery with Normalized Cut

Paper: Wang et al. (2022). "Self-Supervised Transformers for Unsupervised Object Discovery using Normalized Cut."

Link: <https://inria.hal.science/hal-03585410/file/TokenCut.pdf>

arXiv: <https://arxiv.org/html/2209.00383v3>

TokenCut uses DINO features to build a graph. Each image patch is a node. Edges connect patches with similar features. The method solves a Normalized Cut problem. The cut separates foreground from background.

```
DINO patch features
  |
  v
+-----+
| Build similarity |
| graph           |
| (cosine sim)    |
+-----+
  |
  v
+-----+
| Normalized Cut  | (second eigenvalue)
+-----+
  |
  v
+-----+
| Foreground mask |
+-----+
```

Results: CorLoc 68.8% on VOC07, 72.1% on VOC12, 58.8% on COCO20K.

Relevance: The graph method finds structures without labels. It can find the otolith foreground. It may also find increment regions if applied at a finer scale.

4.4 FOUND: Unsupervised Object Localization

Paper: Simeoni et al. (2023). "Unsupervised Object Localization: Observing the Background To Discover Objects."

Link: https://openaccess.thecvf.com/content/CVPR2023/papers/Simeoni_Unsupervised_Object_Localization_Observing_the_Background_To_Discover_Objects_CVPR_2023_paper.pdf

FOUND discovers the background first. It finds the background seed patch. The seed patch has low attention. Then it finds patches similar to the seed. These patches are the background. The complement is the foreground.

The model uses a single 1x1 convolution. It has only 770 trainable parameters. The DINO backbone stays frozen.

```
DINO ViT-S/8 (frozen)
  |
```


Results: The method beats point-level supervised methods on PASCAL VOC 2007 (mRMSE 0.29 vs 0.31) and COCO (0.34 vs 0.35).

Relevance: The density head learns spatial locations from count labels only. This is the same principle as the otolith density head. The age is the count. The density map shows where the increments are.

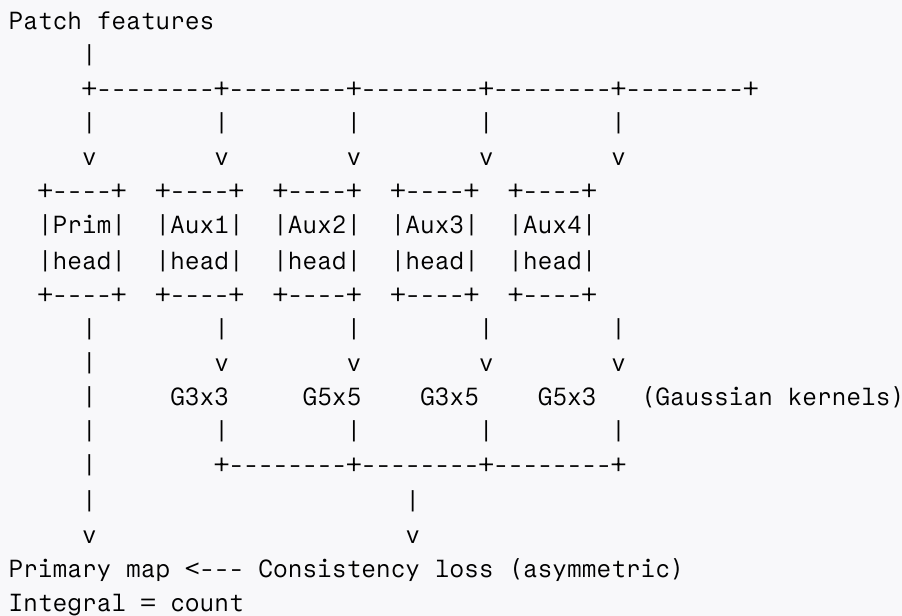
4.6 MATT: Count-Level Weak Supervision with Multi-Branch Consistency

Paper: Liu et al. (2020). "Towards Using Count-level Weak Supervision for Crowd Counting."

Link: <https://arxiv.org/pdf/2003.00164v1>

MATT uses a small set with point labels and a large set with count labels. The method has one primary density head and four auxiliary heads. The auxiliary heads use different Gaussian kernels. A consistency loss makes the auxiliary maps agree with the primary map.

The training is asymmetric. Gradients from the auxiliary heads do not flow into the primary head. The primary head is treated as fixed during the consistency loss.



Results: On ShanghaiTech Part A, MAE 80.1. On MSCC (15 object categories), average MAE 6.8. This is a 63.8% reduction from the baseline.

Relevance: MATT shows that a small annotated set plus a large count-only set works well. The multi-branch consistency prevents the density map from collapsing. The same approach can train the otolith density head.

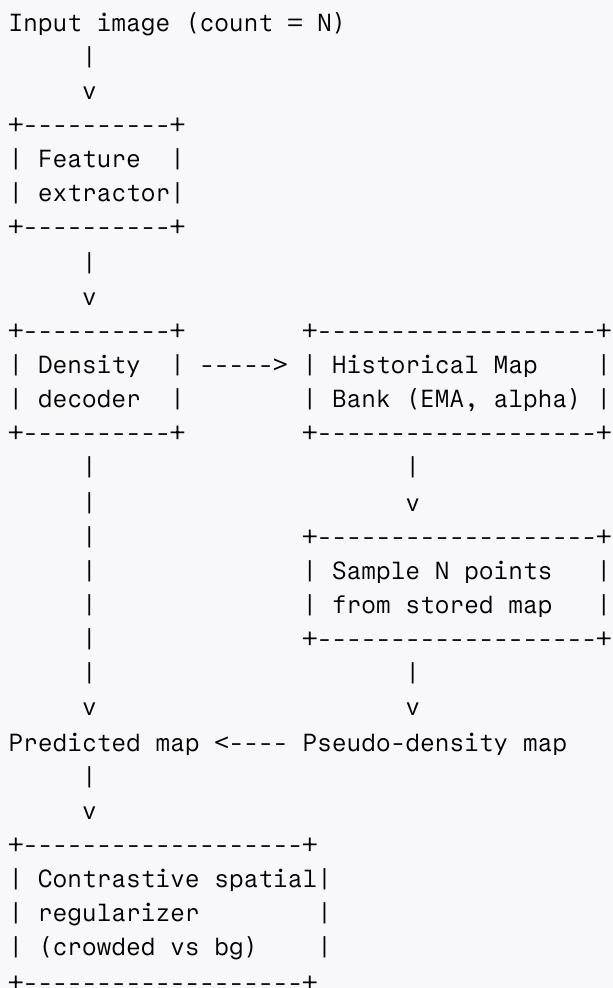
4.7 Count2Density: Density Maps from Count Labels Only

Paper: "Count2Density: Crowd Density Estimation without Location-level Annotations." 2025.

Link: <https://arxiv.org/pdf/2509.03170>

Count2Density trains a density map from count labels only. No point or pixel labels are used. The method keeps a Historical Map Bank. The bank stores an exponential moving average of predicted maps. The bank updates each epoch.

The method uses a contrastive spatial regularizer. It divides the image into crowded and background regions. It makes positive pairs from crowded regions. It makes negative pairs from crowded and background regions.



Results: F1 71.3% on point localization. SSIM 0.85 on density map quality. These results are close to fully supervised methods.

Relevance: Count2Density shows that count labels alone can train a useful density map. The Historical Map Bank gives the model a spatial prior. The contrastive regularizer prevents the map from becoming uniform. The otolith age can serve as the count.

4.8 Additive MIL: Interpretable Patch-Level Heatmaps

Paper: Javed et al. (2022). "Additive MIL: Intrinsically Interpretable Multiple Instance Learning for Pathology."

Link: <https://arxiv.org/html/2206.01794>

This method comes from histopathology. It treats a whole slide image as a bag of patches. The slide label is the bag label. The model learns attention weights for each patch. The attention weights show which patches are important.

The additive decomposition computes a per-patch contribution. Each patch gets a class-wise score. The sum of all patch scores gives the slide prediction. The per-patch scores make a heatmap.

```
Otolith image (bag of patches)
  |
  v
+-----+
| Feature | (DINO, frozen)
| extractor|
+-----+
  |
  v
+-----+
| Attention| (alpha_i = softmax)
| MLP      |
+-----+
  |
  v
+-----+
| Predictor| (psi_p, per-patch)
+-----+
  |
  v
Patch contributions c_i (per class)
  |
  +-----> Sum ----> Bag prediction (age)
  |
  +-----> Sigmoid ----> Heatmap per class
```

Results: AUPRC 0.42 on Camelyon16 cancer localization. A pathologist preferred the additive heatmaps in 49 of 50 slides.

Relevance: The otolith is a bag of patches. The age is the bag label. The model learns which patches are growth increments. The heatmap shows the increment locations. This is directly analogous to the otolith problem.

4.9 CORAL: Rank-Consistent Ordinal Regression

Paper: Cao, Mirjalili, Raschka (2020). "Rank Consistent Ordinal Regression for Neural Networks with Application to Age Estimation."

Link: <https://arxiv.org/abs/1901.07884>

PDF: <https://sebastianraschka.com/pdf/papers/2020-coral.pdf>

Code: <https://github.com/Raschka-research-group/coral-pytorch>

CORAL transforms ordinal targets into binary classification tasks. Each binary task asks: "Is the age greater than k?" The method ensures rank consistency. If the model says age > 5, it must also say age > 4.

The method is architecture-agnostic. You can add it to any backbone. The loss is binary cross-entropy on each binary task.

```
Input: [CLS] token (D-dim)
  |
  v
+-----+
| Shared fully |
| connected layer |
+-----+
  |
  v
+-----+
| K-1 binary   |   Each output: "Is age > k?"
| classifiers   |   Weight matrix: shared, bias: independent
+-----+
  |
  v
Age prediction (sum of probabilities)
```

Relevance: The otolith age is an ordinal variable. Age 5 is one step above age 4. CORAL respects this order. Standard cross-entropy does not. CORAL is the correct loss for the age head.

4.10 Otolith Deep Learning Literature

4.10.1 Automatic Interpretation of Otoliths

Paper: Moen et al. (2018). "Automatic interpretation of otoliths using deep learning."

Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6296523/>

This work uses Inception v3 to predict age from otolith images. The model uses MSE loss for regression. The model does not localize growth increments. It predicts age only.

Results: MSE 2.65 for paired otoliths. Exact age in 29% of test pairs. Within one year in 67% of pairs.

Gap: The model does not show where the increments are. It is a black box.

4.10.2 Annotating Otoliths with a Deep Generative Model

Paper: Bojesen et al. (2024). "Annotating otoliths with a deep generative model."

Link: <https://academic.oup.com/icesjms/article/81/1/55/7341839>

This model predicts the next increment point on an otolith. It uses point annotations for training. The model is autoregressive. It places one point at a time.

Results: Expert quality score 4.22 out of 5. Human score was 4.33. Age agreement 64% to 69%.

Gap: The model needs point annotations for training. It cannot learn from age labels alone.

4.10.3 Fish Age Reading with Object Detection

Paper: Cayetano et al. (2024). "Fish age reading using deep learning methods for object-detection and segmentation."

Link: <https://academic.oup.com/icesjms/advance-article/doi/10.1093/icesjms/fsae020/7614790>

This work uses Mask R-CNN to detect growth rings. It needs full mask annotations. The annotations are labor-intensive.

Results: 46% accuracy on North Sea images. 72% on Baltic Sea images.

Gap: The method needs pixel-level annotations. It cannot use weak supervision.

5. Method Variants

This section describes eight training variants for the density head. Each variant uses a different approach from the literature.

5.1 Variant 1: Baseline (Density Integral Constraint)

Literature: Cholakal et al. (2019). [https://openaccess.thecvf.com/content_CVPR_2019/papers/Cholakka l Object Counting and Instance Segmentation With Image-Level Supervision CVPR 2019 paper.pdf](https://openaccess.thecvf.com/content_CVPR_2019/papers/Cholakka%20Object%20Counting%20and%20Instance%20Segmentation%20With%20Image-Level%20Supervision_CVPR_2019_paper.pdf)

Supervision: Age labels only (count-level).

How it works: The density head makes a non-negative map. The sum of the map must equal the age. The loss is the MSE between the map sum and the age.

```
Patch features [B, N, D]
|
v
+-----+
| Linear(D, 1) |
| + ReLU       |
+-----+
|
v
Density map [B, H, W]
|
```

```

      v
sum(dim=(1,2)) ----> MSE ----> age label

```

Loss:

```
loss = CORAL_loss + lambda * MSE(sum(density_map), age)
```

Expected failure mode: The map may collapse to one point. The model can satisfy the sum constraint without correct spatial placement.

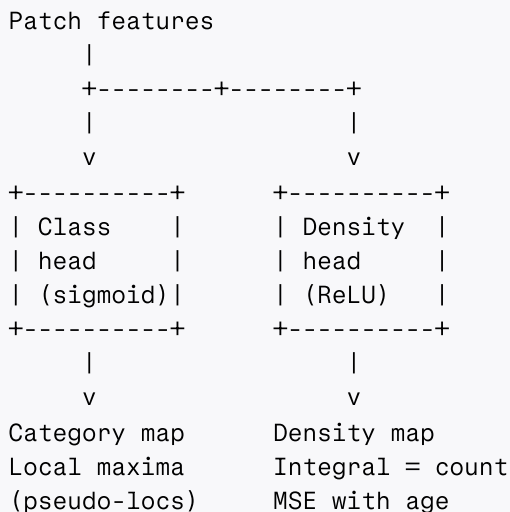
Expert evaluation: Use the expert test set. Compare the map peaks to the expert points. Compute point distance and IoU.

5.2 Variant 2: Cholakal (Classification + Density Heads)

Literature: Cholakal et al. (2019). https://openaccess.thecvf.com/content_CVPR_2019/papers/Cholakal_Object_Counting_and_Instance_Segmentation_With_Image-Level_Supervision_CVPR_2019_paper.pdf

Supervision: Age labels with count in range 1 to 4. Ranking loss for counts above 4.

How it works: The classification head makes a category map. Local maxima on this map give pseudo-locations. The density head learns from these pseudo-locations. The integral of the density map gives the count.



Loss:

```

loss = CORAL_loss
      + lambda_class * classification_loss
      + lambda_density * density_point_loss
      + lambda_density * density_integral_loss
      + lambda_rank * ranking_loss

```

Expected failure mode: The local maxima may not correspond to real increments. The pseudo-locations can be wrong.

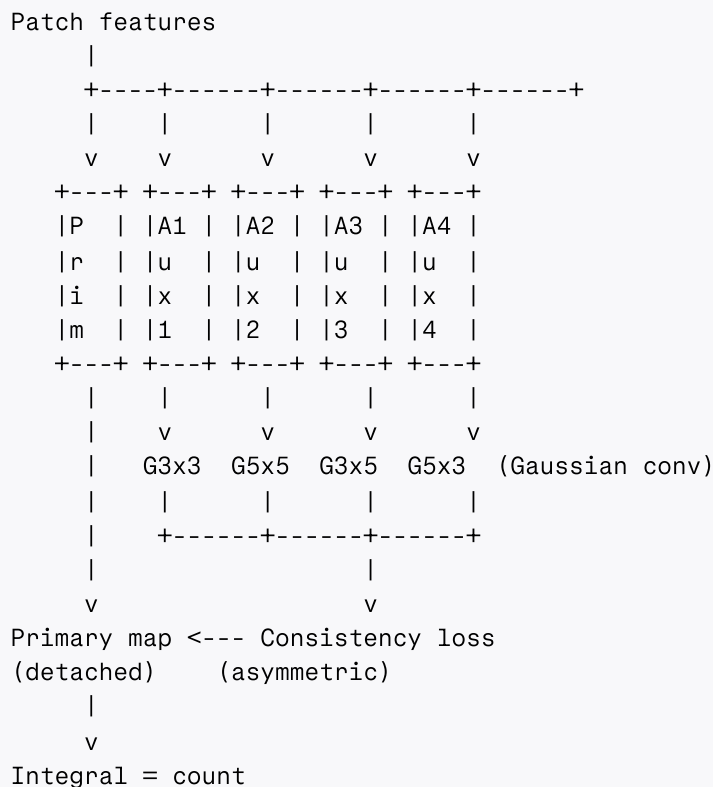
Expert evaluation: Check if the local maxima match the expert points. Compute precision and recall of the maxima.

5.3 Variant 3: MATT (Multi-Branch Auxiliary Consistency)

Literature: Liu et al. (2020). <https://arxiv.org/pdf/2003.00164v1>

Supervision: Small set with point labels + large set with count labels.

How it works: The primary density head makes the main map. Four auxiliary heads make alternative maps. Each auxiliary map uses a different Gaussian kernel. A consistency loss makes the auxiliary maps agree with the primary map. The training is asymmetric. Gradients from auxiliary heads do not update the primary head.



Loss:

```

loss = CORAL_loss
      + lambda_primary * MSE(sum(primary), age)
      + lambda_aux * sum(MSE(sum(aux_i), age) + consistency_loss_i)

```

Expected failure mode: The auxiliary heads may agree on a wrong pattern. The consistency loss can reinforce errors.

Expert evaluation: Compare the primary map to the expert density map. The multi-branch approach should give smoother maps. Check IoU and F1.

5.4 Variant 4: Count2Density (Historical Map Bank + Contrastive)

Literature: Count2Density (2025). <https://arxiv.org/pdf/2509.03170>

Supervision: Count labels only (age).

How it works: The model keeps a Historical Map Bank. The bank stores an EMA of past predictions. Each epoch, the bank updates. The method samples N points from the stored map. N is the age. The sampled points make a pseudo-density map. The model learns from this pseudo-map. A contrastive loss separates crowded and background regions.

```
Training epoch:
  |
  v
+-----+
| Predict density |
| map for each image|
+-----+
  |
  v
+-----+
| Update Historical | H = alpha * pred + (1-alpha) * H
| Map Bank (EMA)   |
+-----+
  |
  v
+-----+
| Normalize stored | Divide by max, smooth with Gaussian
| map              |
+-----+
  |
  v
+-----+
| Sample N points  | N = age label
| from normalized  |
| map              |
+-----+
  |
  v
+-----+
| Make pseudo-density|
| map from points    |
+-----+
  |
  v
+-----+
| Map loss +        |
| Contrastive loss + |
```

```
| Count loss |
+-----+
```

Loss:

```
loss = CORAL_loss
      + lambda_map * MSE(predicted, pseudo_density)
      + lambda_contrast * InfoNCE(crowded, background)
      + lambda_count * MSE(sum(predicted), age)
```

Expected failure mode: The Historical Map Bank can reinforce a wrong pattern. The self-training loop may not converge.

Expert evaluation: Check if the crowded regions match the expert increment locations. Compute precision and recall.

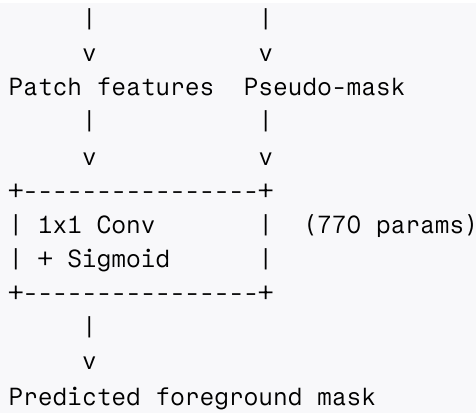
5.5 Variant 5: FOUND (DINO Attention Pseudo-Masks)

Literature: Simeoni et al. (2023). https://openaccess.thecvf.com/content/CVPR2023/papers/Simeoni_Unsupervised_Object_Localization_Observing_the_Background_To_Discover_Objects_CVPR_2023_paper.pdf

Supervision: None. Uses DINO self-attention only.

How it works: The method finds the background first. It selects a seed patch with low attention. It finds patches similar to the seed. These are the background. The complement is the foreground. A single 1x1 convolution learns to predict the foreground. The training uses pseudo-masks from the background discovery. After 100 iterations, the model uses its own predictions as targets.

```
DINO ViT-S/8 (frozen)
|
+---> Self-attention maps
|
|          |
|          v
|  +-----+
|  | Find background |
|  | seed patch      |
|  | (low attention) |
|  +-----+
|          |
|          v
|  +-----+
|  | Find similar    |
|  | patches (bg)    |
|  +-----+
|          |
|          v
|  +-----+
|  | Foreground =    |
|  | 1 - background  |
|  +-----+
```



Loss:

```

# Phase 1 (first 100 iterations):
loss = lambda_bg * BCE(predicted, refined_background_mask)

# Phase 2 (after 100 iterations):
loss = lambda_self * BCE(predicted, self_prediction) # if IoU > 0.5

```

Expected failure mode: The method may find the otolith outline. It may not find individual increments. The background discovery is for objects, not for periodic structures.

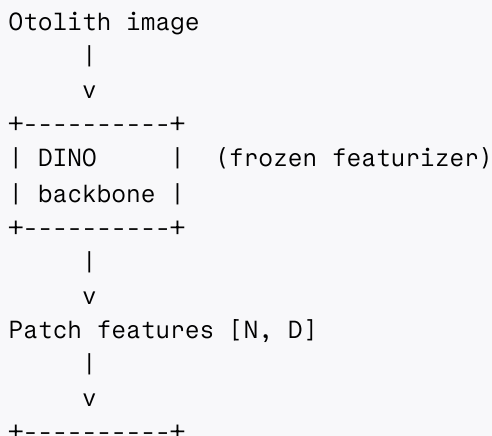
Expert evaluation: Check if the foreground mask covers the increment regions. The mask may be too coarse for individual increments.

5.6 Variant 6: Additive MIL (Attention-Based Multiple Instance Learning)

Literature: Javed et al. (2022). <https://arxiv.org/html/2206.01794>

Supervision: Image-level labels (age).

How it works: The otolith is a bag of patches. The age is the bag label. An attention MLP computes a weight for each patch. The weighted patch features go to a predictor. The predictor makes a per-patch contribution. The sum of all contributions gives the age prediction. The per-patch contributions make the heatmap.



```

| Attention|  alpha_i = softmax(MLP(features_i))
| MLP      |
+-----+
|
v
Weighted features: m_i = alpha_i * features_i
|
v
+-----+
| Predictor|  c_i = psi_p(m_i)  [per patch, per class]
| MLP      |
+-----+
|
+-----> Sum(c_i) ----> Age prediction
|
+-----> Sigmoid(c_i) ----> Heatmap per class

```

Loss:

```

loss = CORAL_loss
      + lambda_bag * cross_entropy(bag_prediction, age)
      + lambda_density * MSE(sum(heatmap), age)
      + lambda_sparsity * entropy(attention_weights)

```

Expected failure mode: The attention may focus on one dominant patch. The entropy regularizer must prevent this. The heatmap may be too diffuse.

Expert evaluation: Compare the heatmap to the expert density map. Check if high-attention patches contain increments. Compute IoU after thresholding.

5.7 Variant 7: TokenCut (Graph-Based Foreground Discovery)

Literature: Wang et al. (2022). <https://inria.hal.science/hal-03585410/file/TokenCut.pdf>

Supervision: None. Uses DINO features only.

How it works: The method builds a graph from DINO patch features. Each patch is a node. Edges connect patches with cosine similarity above a threshold. The method solves the Normalized Cut problem. It finds the second smallest eigenvalue of the graph Laplacian. The eigenvector splits the image into foreground and background.

```

DINO patch features [N, D]
|
v
+-----+
| Cosine similarity |
| matrix [N, N]    |
| threshold tau=0.2 |
+-----+
|
v

```

```

+-----+
| Adjacency matrix |
| E (binary)      |
+-----+
|
| v
+-----+
| Degree matrix D |
| Laplacian L=D-E |
+-----+
|
| v
+-----+
| Eigenvalue      |
| decomposition   |
| 2nd smallest    |
+-----+
|
| v
+-----+
| Fiedler vector  |
| split at mean   |
+-----+
|
| v
Foreground / Background mask

```

Loss: The TokenCut step has no loss. It is a graph algorithm. The density head learns from the TokenCut output:

```

loss = CORAL_loss
      + lambda_tokencut * MSE(density_head_output, tokencut_mask)
      + lambda_count * MSE(sum(density), age)

```

Expected failure mode: TokenCut finds one main foreground object. It may find the otolith but not individual increments. For periodic structures, the graph may not separate increments from gaps.

Expert evaluation: Check if the foreground mask overlaps with increment regions. The mask may be too coarse for individual increment detection.

5.8 Variant 8: Semi-Supervised (Small Annotated Set + Large Age-Only Set)

Literature:

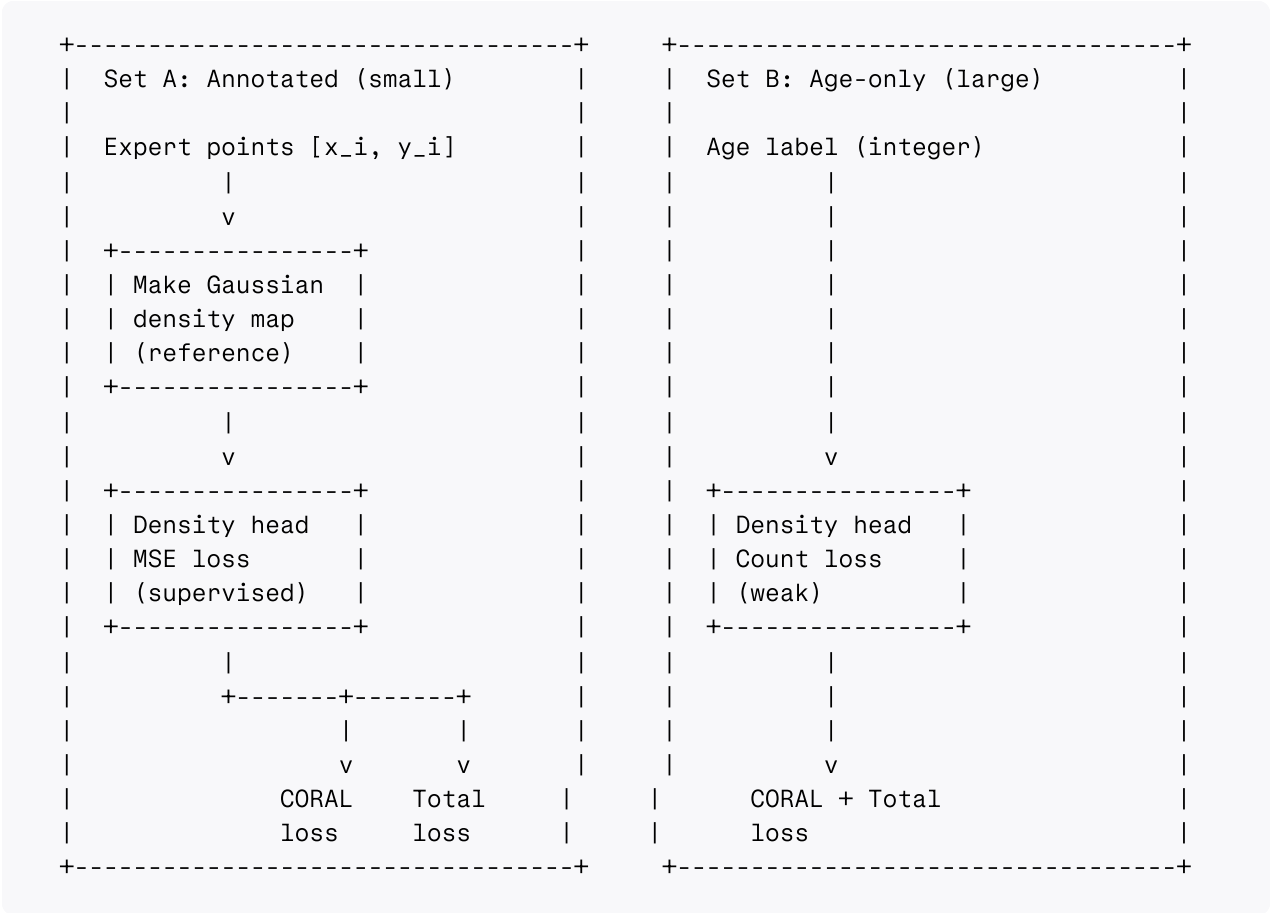
- Liu et al. (2020). MATT. <https://arxiv.org/pdf/2003.00164v1>
- Cholakal et al. (2019). https://openaccess.thecvf.com/content_CVPR_2019/papers/Cholakal_Object_Counting_and_Instance_Segmentation_With_Image-Level_Supervision_CVPR_2019_paper.pdf
- Bojesen et al. (2024). <https://academic.oup.com/icesjms/article/81/1/55/7341839>

Supervision: Small set with expert point labels + large set with age labels.

How it works: The system uses two data sets. Set A has 20 to 50 images. Each image has expert points that mark the increments. Set B has 1000+ images. Each image has an age label only.

The system makes a reference density map from the expert points. Each point becomes a Gaussian blob. The sum of all blobs is the reference map.

The density head trains on both sets. On Set A, the loss is the MSE between the predicted map and the reference map. On Set B, the loss is the MSE between the map sum and the age.



Loss:

```
# Set A (annotated):
loss_a = CORAL_loss
        + lambda_supervised * MSE(density_map, reference_density_map)
        + lambda_count_a * MSE(sum(density_map), age)

# Set B (age-only):
loss_b = CORAL_loss
        + lambda_count_b * MSE(sum(density_map), age)

# Total:
loss = loss_a + loss_b
```

Training Strategies:

Strategy A: Alternating

=====

Epoch loop:

1. Train one batch from Set A (supervised)
2. Train one batch from Set B (weak)
3. Repeat

Strategy B: Mixed Batching

=====

Epoch loop:

1. Make a batch with 20% from Set A and 80% from Set B
2. Compute supervised loss for Set A images
3. Compute count loss for Set B images
4. Backpropagate combined loss
5. Repeat

Strategy C: Pre-train then Finetune

=====

Phase 1:

Train density head on Set A only (50 epochs)
Use supervised density loss

Phase 2:

Finetune on Set A + Set B (100 epochs)
Use supervised + count loss

Strategy D: Pseudo-Label Propagation

=====

Phase 1:

Pre-train on Set A (supervised)

Phase 2:

Predict density maps for Set B
Store predictions as pseudo-labels

Phase 3:

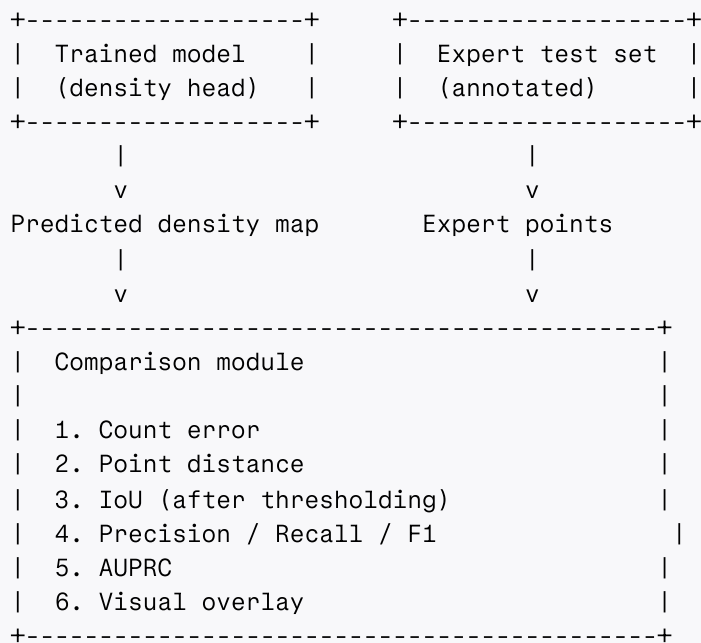
Train on Set A (hard labels) + Set B (pseudo-labels)
Weight Set B loss by confidence

Expected failure mode: The model may overfit to Set A if Set A is too small. The pre-train strategy may help. The pseudo-label strategy may reinforce errors if the initial model is poor.

Expert evaluation: This variant has the strongest evaluation. Use a separate test set with expert annotations. Compute IoU, precision, recall, F1, and AUPRC. Compare to all other variants.

6. Evaluation Methodology

6.1 Overview



6.2 Count Error

The count error checks the total. The sum of the density map should equal the age.

```
count_error = abs(sum(density_map) - true_age)
relative_count_error = count_error / true_age
```

6.3 Point Distance

The point distance checks spatial accuracy. Find the peaks on the density map. Each peak should match an expert point.

Expert points (marked by human)	Density map peaks (model prediction)	
* ---- 3px ---- *	X ---- 1px ---- X	Good match
* ---- 10px --- *	X ---- 15px --- X	Poor match

For each expert point:
Find the nearest predicted peak
Record the distance

mean_distance = average of all distances

6.4 IoU (Intersection over Union)

The IoU checks region overlap. Threshold the density map. Compare the binary mask to the expert mask.

```
Expert mask          Predicted mask          Overlap (IoU)
+----+----+         +----+----+         +----+----+
|###|      |         |###|###|         |###|      |
|###|      |         |###|###|         |###|      |
+----+----+         +----+----+         +----+----+
      Intersection = ##      Union = #####
      IoU = Intersection / Union
```

6.5 Precision, Recall, F1

```
Predicted increments
+----+----+----+
| TP | FP | FN |      TP = True Positive (correct)
+----+----+----+      FP = False Positive (wrong)
+----+----+----+
| TP | TP | FN |      FN = False Negative (missed)
+----+----+----+

Precision = TP / (TP + FP)    "How many predictions are correct?"
Recall = TP / (TP + FN)       "How many expert points did we find?"
F1 = 2 * P * R / (P + R)     "Balance of both"
```

6.6 AUPRC (Area Under Precision-Recall Curve)

The AUPRC checks the model at all thresholds. Sweep the threshold from 0 to 1. At each threshold, compute precision and recall. Plot the curve. The area under the curve is the AUPRC.

6.7 Visual Overlay

The visual overlay shows the density map on top of the otolith image. The expert points are also shown. This lets a human judge the quality.

```
Otolith image
|
v
+-----+
| Overlay: |
| - Otolith photo |
| - Density map   | (semi-transparent heat colors)
| - Expert points | (white dots)
+-----+
```

7. Comparison Table

Variant	Supervision	Can we test localization?	Literature	Key idea
1. Baseline	Age only	No (no spatial signal)	Cholakkal 2019	Map sum = age
2. Cholakkal	Age + count range	No (pseudo-locations)	Cholakkal 2019	Class head gives pseudo-locs
3. MATT	Small points + large count	No (no direct eval)	Liu 2020	Multi-branch consistency
4. Count2Density	Count only	No (self-generated)	2025	Historical Map Bank
5. FOUND	None	No (unsupervised)	Simeoni 2023	DINO attention pseudo-masks
6. Additive MIL	Image-level	No (no direct eval)	Javed 2022	Per-patch attention from bag label
7. TokenCut	None	No (unsupervised)	Wang 2022	Normalized Cut on DINO graph
8. Semi-supervised	Small expert + large age	YES (direct comparison)	Liu 2020 , Cholakkal 2019	Expert labels train + test

8. Recommended Experiment Plan

8.1 Step 1: Prepare Data

1. Select 50 otolith images for Set A.
2. Ask an expert to mark growth increments on each image.
3. Store the points as (x, y) coordinates.
4. Store the age for each image.
5. Select 1000+ otolith images for Set B.
6. Store the age for each image.
7. Reserve 20 images from Set A as the expert test set.

8.2 Step 2: Run All Variants

1. Run Variant 1 (Baseline) as the reference.
2. Run Variants 2 to 7 in order.
3. Run Variant 8 (Semi-supervised) last.
4. For Variant 8, test Set A sizes: 5, 10, 20, 30, 50.
5. For Variant 8, test all four training strategies.

8.3 Step 3: Evaluate

1. For each variant, predict the density map on the expert test set.
2. Compute count error, point distance, IoU, precision, recall, F1, AUPRC.
3. Make visual overlays for each variant.
4. Record all results in a table.

8.4 Step 4: Compare

1. Compare all variants on count error and age MAE.
2. Compare all variants on localization metrics (IoU, F1, AUPRC).
3. Identify the best variant for age prediction.
4. Identify the best variant for localization.
5. Check if the best age variant is also the best localization variant.

8.5 Step 5: Ablation

1. For Variant 8, compare Set A sizes: 5, 10, 20, 30, 50, 100.
2. For each size, run all four training strategies.
3. Plot the localization metrics against the Set A size.
4. Find the minimum Set A size for acceptable localization.

9. Key Findings from Literature

9.1 DINO Features Show Structures Without Labels

The DINO backbone makes attention maps that show object parts. This happens without any labels. The paper by Caron et al. (2021) shows this result. The features are available for the otolith project. The backbone does not need retraining.

Link: <https://arxiv.org/abs/2104.14294>

9.2 Count Labels Can Train Density Maps

Cholakkal et al. (2019) and Count2Density (2025) show this. A density map can learn from count labels only. The integral of the map equals the count. The map shows spatial locations.

Links:

- https://openaccess.thecvf.com/content_CVPR_2019/papers/Cholakkal_Object_Counting_and_Instance_Segmentation_With_Image-Level_Supervision_CVPR_2019_paper.pdf
- <https://arxiv.org/pdf/2509.03170>

9.3 A Small Annotated Set Improves Results

MATT (Liu et al., 2020) shows this. A small set with point labels plus a large set with count labels gives good results. The small set provides spatial supervision. The large set prevents overfitting.

Link: <https://arxiv.org/pdf/2003.00164v1>

9.4 Expert Annotations Enable Localization Testing

No paper in the literature tests density head localization on otoliths with expert annotations. This is a gap. The small annotated set in this project fills the gap. It makes the localization measurable. You can compare the model output to expert marks.

9.5 CORAL Is the Correct Loss for Age

The otolith age is an ordinal variable. CORAL (Cao et al., 2020) respects the order. Standard cross-entropy does not. CORAL is the correct loss for the age head.

Link: <https://arxiv.org/abs/1901.07884>

9.6 Existing Otolith Methods Do Not Test Localization

The otolith literature uses black-box models or needs full annotations. Moen et al. (2018) predict age without localization. Bojesen et al. (2024) need point annotations. Cayetano et al. (2024) need mask annotations. No paper tests localization with weak supervision.

Links:

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC6296523/>
- <https://academic.oup.com/icesjms/article/81/1/55/7341839>
- <https://academic.oup.com/icesjms/advance-article/doi/10.1093/icesjms/fsae020/7614790>

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12. Bojesen et al. (2024). "Annotating otoliths with a deep generative model." <https://academic.oup.com/icesjms/article/81/1/55/7341839>
13. Cayetano et al. (2024). "Fish age reading using deep learning methods for object-detection and segmentation." <https://academic.oup.com/icesjms/advance-article/doi/10.1093/icesjms/fsae020/7614790>